

**First Impressions in Online Dating: Effects of Message Type and Form on Perceptions and Intention
to Respond Among Dutch Young Adults**

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Abstract

This study investigates how different types of pick-up lines and message form influence perceptions of the receiver and the intention to respond in online dating apps. Fifty-seven Dutch individuals aged 18 to 34 participated in a 3 (type: flippant, direct complimentary, and innocuous) $\times$ 2 (form: question versus no question) online experiment. Responses were analyzed using two-way repeated measures ANOVAs and mediation analyses, with gender differences observed as well. Results showed a significant effect of pick-up line type on perceptions and intention to respond, with direct complimentary lines rated highest for sociability, romantic attraction, trustworthiness, intelligence, and intention to respond. Mediation analyses revealed that intention to respond was primarily mediated by perceptions of sociability and romantic attraction. Statements enhanced romantic attraction more than questions, possibly due to the confidence they convey. Significant interaction effects between pick-up line type and form suggest that the influence of the form on perceived humor and romantic attraction varies depending on the type. Gender differences emerged with women rating direct complimentary lines lower in romantic attraction and trustworthiness than men, though traits like humor and intelligence are perceived similarly across genders. The findings emphasize the importance of confident and effective communication in online dating. Limitations include the absence of profile pictures and usernames, limited message stimuli variation, and potential carryover effects. Findings offer practical insights for dating app users and developers to improve messaging strategies and provide insights into the user dynamics in online dating contexts.

Keywords: Tinder, online dating, pick-up lines, humor, creativity, sociability, romantic attraction, trustworthiness, intelligence.

First Impressions in Online Dating: Effects of Message Type and Form on Perceptions and Intention to Respond Among Dutch Young Adults

The dating scene has changed noticeably in recent years as more people begin to look for a partner online (Hobbs et al., 2017). According to Harford (2016), finding a romantic partner is considered to be one of the biggest challenges individuals face, and the rise of online dating represents an important innovation that makes this process easier. This shift to online dating introduces new dynamics in initial interactions, raising questions about how individuals initiate conversations and what makes these interactions successful.

Nowadays, people have access to an enormous number of potential partners (Pronk & Denissen, 2020). For example, users of the popular dating app Tinder see an average of 140 possible matches every day (Smith, 2018). While the abundance of available matches is a major reason for the success of online dating apps (Lenton & Stewart, 2008), it also creates challenges for users to stand out in an oversaturated market. Individuals tend to favor having numerous options, which should increase the likelihood of finding a match (Lancaster, 1990; Patall et al., 2008), but it can also cause people to feel overwhelmed (Pronk & Denissen, 2020). Nevertheless, many people meet their partners online; in 2017, 39% of heterosexual adults in the United States met their partners through online platforms (Rosenfeld et al., 2019).

An abundance of choices

In the competitive world of online dating, many initial messages remain unanswered (Kallis, 2017), reflecting the challenge of standing out on dating apps. Research indicates men receive replies to about half of their messages, while women face even lower response rates (Zhang & Yasseri, 2016). The lack of responses can be attributed to the overwhelming number of potential matches available (Best & Delmege, 2012), which increases the likelihood of rejection (Van de Wiele & Campbell, 2019). Unlike traditional dating, where face-to-face interactions naturally create expectations for involvement in a conversation, online dating allows individuals to ignore interactions (Van de Wiele & Campbell, 2019). The anonymity of online interactions makes rejection easier, as users can ignore messages without facing the social consequences that might occur in offline settings (Hitsch et al., 2010). In addition to the negative effect rejection has on people's self-image, it can contribute to an increased risk of depression, anxiety, and distress (Holtzhausen et al., 2020). Consequently, the first impression in an online dating chat becomes even more important to ensure that a response is received at all.

The importance of initial messages in online dating

The initiation of a conversation plays a critical role in any interaction (Schegloff, 1987), particularly on online dating apps, where first impressions are often formed quickly through short textual exchanges (Van Der Zanden et al., 2019). Sending an effective first message is essential for increasing the chances of a response and determining whether a relationship will continue (Fisher et al., 2020). The Social Information Processing Theory suggests that when certain social cues, such as facial expressions or body language, are absent, individuals tend to depend more on the cues that are available (Walther, 1992). Therefore, users of online dating apps focus more on textual elements (Walther, 1992), amplifying their impact on how impressions are formed.

Text often contains implicit information (Tausczik & Pennebaker, 2010), such as tone or specific word choice, which plays a crucial role in conveying signals about potential matches. These cues can be both unintentional and intentional, aiming to communicate specific messages (Goffman, 1959). While intentional cues are deliberately crafted, unintentional cues emerge through several factors, including the length of the message (Schoendienst & Dang-Xuan, 2011), typographic features (Lea & Spears, 1992), and language errors (Van Der Zanden et al., 2019). Research by Ellison et al. (2006) shows that linguistic features of messages, such as timing, length, and grammar, can be as important as the content of the initial interaction itself in online dating conversations. Schoendienst and Dang-Xuan (2011) further emphasize that individuals are less likely to respond to an initial message if the sender uses more self-references. Instead, people tend to favor messages that focus on them, thereby increasing the likelihood of eliciting engagement (Schoendienst & Dang-Xuan, 2011). Additionally, women are more likely to respond to longer messages, while men prefer short messages (Schoendienst & Dang-Xuan, 2011). This gender difference implies that men should invest effort in crafting more detailed messages, while women should keep their communications shorter if they want to be more successful with their opposite sex (Schoendienst & Dang-Xuan, 2011). Another study by Van Der Zanden et al. (2019) demonstrated that initial messages with linguistic errors related to mechanics and grammar reduced perceptions of attentiveness and intelligence, thereby decreasing attractiveness and romantic interest. Although these studies emphasize the importance of implicit textual factors in online dating, no research has examined whether the form of a message, such as containing a question, affects responses.

The influence of message form

Unintentional cues contribute to forming a more realistic perception of the person as they are not consciously created (Walther & Parks, 2002; Wotipka & High, 2016). In the context of dating profiles, these cues are seen as more critical for forming an accurate impression of the sender due to their high degree of perceived trustworthiness (Ellison et al., 2006). One key aspect of unintentional

cues is the form of the pick-up line. This study specifically focuses on the distinction between pick-up lines that are framed as a question and those that do not contain a question. Research by Dillon (1982) suggests that this influences how a message is received and interpreted, thereby influencing the response. Questions encourage interaction by inviting the recipient to respond, which initiates a conversation (Dillon, 1982). This sense of reciprocity, where a question prompts a response, may increase the likelihood of receiving a reply (Dillon, 1982). In the context of online dating, where many potential matches remain unengaged (Kallis, 2017), asking questions may act as an effective way to start a conversation. While research shows that questions promote interaction, less is known about how this works within the context of online dating.

The rise of Tinder

Online dating primarily occurs through apps, each offering unique approaches to connect users with potential matches by featuring concise profiles and utilizing filters like age, gender, and location (Koivisto et al., 2023). Tinder rapidly emerged as the largest online dating app in Western society (Duguay, 2016), with about 1.5 million users in the Netherlands by 2014 (Zandstra, 2014). By 2018, Tinder had more than 50 million users worldwide, generating two billion swipes and 30 million matches per day (Iqbal, 2024). Tinder allows users to swipe left or right based on the first impression of a dating profile. When both users swipe to the right, a match is created, and the opportunity to chat with each other arises (Iqbal, 2024). This makes it possible to engage in multiple conversations simultaneously, instead of being limited to one conversation at a time, as in offline settings (LeFebvre, 2017). Individuals aged 18 to 34 are the most frequent users of online dating apps like Tinder (Iqbal, 2024). Despite this young generation and the modern approach of online dating apps, traditional pick-up lines are still used to start a conversation (Sharabi & Dykstra-DeVette, 2019).

Pick-up lines in online dating

The characteristics of pick-up lines are particularly important because of the absence of non-verbal cues (such as tone and facial expressions) in computer-mediated communication, which are essential for conveying emotions and intentions in face-to-face interactions (Burgoon et al., 2021). Without non-verbal cues, the text alone must convey the sender's intentions, making the style and form of a pick-up line crucial in shaping how it is perceived (Burgoon et al., 2021). Opening phrases, also known as pick-up lines, refer to what someone says when initiating romantic communication (Levine et al., 1994). In this study, all initial messages are referred to as pick-up lines, regardless of their style or intent. Pick-up lines serve as conversation starters (Sharabi & Dykstra-DeVette, 2019), but also play a crucial role in setting the stage for a successful match or even relationship (Levine et al., 1994).

The primary goal of using pick-up lines is to get a date, which can contribute to relationship formation and mate selection (Buss & Schmitt, 1993). Previous research of Kleinke et al. (1986) in offline dating settings categorized pick-up lines into three types: flippant, direct complimentary, and innocuous. Flippant pick-up lines are often cliché and use humor to convey flirtation, frequently with a playful or sexual tone. An example of a flippant pick-up line is: *"You must be tired, because you've been running through my mind all day."* (Kleinke et al., 1986). Direct complimentary lines are straightforward attempts to initiate interactions, often expressing sincerity and offering a compliment. An example of such a line is: *"You seem different. I like that."* (Kleinke et al., 1986). Innocuous pick-up lines are more subtle and indirect. These often start conversations with simple messages that conceal the user's interest, thereby reducing the chance of a potential rejection. An example of this approach is: *"I've seen you before. Do you live around here?"* (Kleinke et al., 1986). Thus, each type of pick-up line has distinct characteristics and plays a crucial role in determining the potential success of a relationship (Koeppel et al., 1993; Levine et al., 1994).

Overview of the current research

This research focuses on (1) pick-up line types; flippant, direct complimentary and innocuous, and (2) message form; pick-up lines framed as a question or not. While previous studies examined pick-up lines in offline settings, little research has explored their impact in an online dating context. By investigating how type and form affect the perceptions of the sender and the intention to respond, this study provides valuable insights into online communication in romantic contexts. Besides, potential mediators, including perceived humor, creativity, sociability, romantic attraction, trustworthiness, and intelligence, are examined in relation to the expected effects of pick-up line type on intention to respond. In the realm of online dating apps, where first impressions are often based on brief textual interactions (Van Der Zanden et al., 2019), it is crucial to understand how different types of pick-up lines and the form of a message impact the perception of the sender and the intention to respond. Therefore, the research question is: *"To what extent do different types of pick-up lines (flippant, direct complimentary, and innocuous) and message form (question versus no question) influence the perception of the sender and the intention to respond on online dating apps among Dutch individuals aged 18 to 34, and how is the effect of type on intention to respond mediated by the perception of the sender?"*

The theoretical framework will provide an overview of the key concepts and relevant literature, followed by a description of the research design in the method section. Subsequently, the results will be presented with a reflection on the findings, implications, limitations, and recommendations for future research in the discussion.

Theoretical Framework

The impact of pick-up lines

The type of pick-up line determines both the direction and the outcome of an interaction by guiding the recipient's response and conversation toward a desired topic or action (Levine et al., 1994). Moreover, pick-up lines reveal the sender's intentions and provide subtle cues about personality traits, influencing the receiver's perception and forming an impression of the sender (Heider, 1982). These varying interpretations of first impressions are crucial in the context of online dating apps, where users quickly evaluate many potential partners (Pronk & Denissen, 2020; Van Der Zanden et al., 2019).

According to the Theory of Attribution (Heider, 1982), receivers tend to overattribute characteristics of a message to the sender's personality. For instance, if a message is perceived as creative or unique, the receiver is likely to conclude that the sender possesses these traits, enhancing their overall impression of the sender (Heider, 1982). A previous study by Berkes (2021) shows that the attribution theory is also applicable in an online dating context, where characteristics of pick-up lines influence the perception of the sender's personality. This highlights the importance of sending the right signals, since even small details of a pick-up line have an impact on the impression left. In light of this overattribution, this study will examine various positive perceptions, including humor, creativity, sociability, romantic attraction, trustworthiness, and intelligence, as these are seen as desirable traits in a potential partner (Senko & Fyffe, 2010).

Flippant, direct complimentary and innocuous pick-up lines

Different types of pick-up lines, including flippant, direct complimentary, and innocuous, may employ distinct strategies to capture attention, convey personality traits, and elicit responses. Flippant lines frequently incorporate humor, which plays an important role in eliciting positive emotions (Kleinke et al., 1986; Skalski et al., 2009). Humor refers to the capacity to express or recognize something that provokes laughter and enjoyment (Hackfort et al., 2019), and helps individuals to differentiate themselves from others while capturing attention (Cunningham & Barbee, 2008; Senko & Fyffe, 2010). It plays a crucial role in successful flirting on online dating platforms (Apostolou & Christoforou, 2020), with a good sense of humor being considered highly attractive (Bressler et al., 2006). Humor is an effective strategy to enhance perceived benefits by stimulating positive emotions and promoting more engaged interactions (Thibaut & Kelley, 1959). It also reduces psychological reactance, the resistance individuals feel when their choices appear limited (Skalski et al., 2009). According to the Principle of Liking (Cialdini, 2009), individuals are more likely to connect with people they find likable or nice. Likeability is recognized as an important factor influencing persuasive communication to attract interest and promote interaction in online dating (Orji et al.,

2015) and can be driven by traits like humor (Cialdini, 2009). However, because humor is subjective, what one person finds funny might not appeal to another (Meyer, 2015). This means it can either enhance connections or fail if it doesn't resonate with the receiver.

Research has shown that humor in online dating improves perceptions of traits such as creativity (Miller, 2000). Creativity is the ability to develop original ideas and products or solve problems in new ways (Parkhurst, 1999). While clichéd flippant lines may suggest low creativity (Masden & Edwards, 2015; Whitty, 2007), these lines can still be perceived as creative due to their humor and contextual delivery. In online dating, a lack of creativity can result in negative perceptions toward potential partners (Masden & Edwards, 2015; Whitty, 2007).

Additionally, previous research in offline dating settings by Cunningham (1989), Kleinke and Dean (1990), and Senko & Fyffe (2010) showed that flippant lines were evaluated more sociable and confident compared to direct complimentary and innocuous lines. Sociability refers to the sender's tendency to socially interact with others and the amount of engagement in social activities (Rubin et al. 1989). This trait influences how attractive someone is perceived as a partner (Meier et al., 2010).

Previous research by Wilbur & Campbell (2011) showed that perceptions of humor that emerged from online dating were positively associated with higher romantic attraction. Romantic attraction refers to the extent to which one evaluates someone as a potential romantic partner (Foster et al., 1998). It affects both the likelihood of receiving a response and the intention to pursue a romantic relationship (Rosenburg & Tunney, 2008). This complex perception can be influenced by the interplay of various characteristics, including intelligence and trustworthiness, as well as the style of initial messages (Gao et al., 2017; Senko & Fyffe, 2010).

Flippant lines can show that someone has a good sense of humor, but they may also suggest a lack of trustworthiness, leading to a less attractive impression (Senko & Fyffe, 2010). Research by Levine et al. (1994) and Kleinke and Dean (1990) showed that flippant lines were rated the lowest in terms of trustworthiness in offline dating settings. Trustworthiness indicates a person's credibility and the likelihood of honest self-presentation (Toma, 2010). Trust is the expectation that others will act beneficially, particularly in uncertain situations like online dating, where users risk deception (Toma, 2010).

Flippant lines intended to be humorous may come across as less serious, making the sender seem less intelligent, as humor is not always received positively (Cunningham, 1989). Intelligence encompasses the perception of the sender's cognitive abilities involving reasoning, problem-solving, and abstract thinking (Gottfredson, 1997). Individuals that are perceived as more intelligent are often seen as more attractive partners (Gao et al., 2017).

Based on the findings that flippant pick-up lines positively influence various perceptions, while potentially having negative effects on trustworthiness and intelligence, the following hypothesis is proposed:

H1: Flippant pick-up lines will result in higher perceived (a) humor, (b) creativity, (c) sociability and (d) romantic attraction, but lower perceived (e) trustworthiness and (f) intelligence compared to direct complimentary and innocuous lines.

Direct complimentary pick-up lines convey sympathy and responsibility, traits associated with higher attractiveness (Kleinke & Dean, 1990). In addition, they tend to be perceived as more sociable than innocuous lines (Senko & Fyffe, 2010). However, prior research in offline dating settings has shown no differences between complimentary and innocuous lines in terms of perceived humor, creativity, trustworthiness, and intelligence (Senko & Fyffe, 2010). Therefore, the following hypothesis is proposed:

H2: Direct complimentary pick-up lines will result in higher (a) sociability and (b) romantic attraction compared to innocuous lines.

In previous research by Kleinke and Dean (1990) in offline dating settings, individuals rated the effectiveness of the three types of pick-up lines based on their ability to attract interest. All innocuous lines were rated as most effective, followed by direct complimentary lines, while flippant lines are considered least effective. Since the effectiveness of a pick-up line is linked to the intention to respond (Kleinke et al., 1986), the following hypothesis is proposed:

H3: Flippant pick-up lines will result in lower intentions to respond compared to direct complimentary and innocuous lines, with innocuous lines eliciting the highest intention to respond.

How questions drive conversations

While the type of pick-up line might influence perceptions and the intention to respond, the form, whether presented as a statement or question, might impact the dynamics of the conversation. Questions play a crucial role in gathering information and guiding conversations (Langacker & Chafe, 1972; Dillon, 1982), particularly in first encounters, such as those on online dating apps (Berger & Calabrese, 1975). Although no single structural form or set of characteristics encompasses all types of questions, they share one common function: eliciting a response (Langacker & Chafe, 1972). This functional definition excludes rhetorical questions, which do not require an answer (Kearsley, 1976). In addition to gathering information, asking questions serves to manage impressions by conveying interest and attention, positively influencing perceptions of the sender

(Huang et al., 2017). This behavior conveys a high degree of responsiveness, an interpersonal trait characterized by providing relevant and appropriate responses to the needs of a conversation partner, which fosters sympathy and engagement (Davis, 1982; Miller & Berg, 1984). No significant effects are expected on humor, creativity, romantic attraction, trustworthiness, and intelligence, as no associations between these traits and the use of questions have been identified in prior research. Given the potential for questions to convey greater responsiveness and sociability compared to statements, the following hypothesis is proposed:

H4: Pick-up lines that are framed as questions will result in more positive perceptions of the sender's sociability compared to lines that are framed as statements.

Pick-up lines framed as questions may be more effective in eliciting responses and creating a favorable impression, because they signal interest and a desire to connect with the recipient (Dillon, 1982). Besides, message form influences conversational dynamics, as asking a question tends to direct the conversation by encouraging the receiver to provide a response, which fosters engagement (Dillon, 1982). Based on this reasoning, the following hypothesis is proposed:

H5: Pick-up lines that are framed as questions will result in higher intentions to respond compared to lines that are framed as statements.

The mediating role of perceptions on response intentions

Given that humor enhances engagement (Thibaut & Kelley, 1959), it is essential to investigate its mediating role in the relationship between different types of pick-up lines and the intention to respond. Higher levels of creativity (Masden & Edwards, 2015; Whitty, 2007) and sociability (Huang et al., 2017) result in more positive responses, while romantic attractiveness further influences the intention to move forward (Rosenburg & Tunney, 2008), thereby affecting the likelihood of receiving a response. Trustworthiness encourages engagement by reducing perceived risks of deception (Toma, 2010), with messages from trustworthy sources being more likely to prompt engagement (Finkel et al., 2012). Lastly, individuals may be more likely to engage with a message if they perceive the sender as intelligent (Gao et al., 2017). Based on these findings, the following is expected:

H6: The effects of different types of pick-up lines on the intention to respond are mediated by perceptions of the sender, with pick-up lines rated high in (a) humor, (b) creativity, (c) sociability, (d) romantic attraction, (e) trustworthiness, and (f) intelligence resulting in a higher intention to respond.

Interaction effect

As we expect pick-up line type and form to independently influence perceptions of the sender and the intentions to respond, it is interesting to explore their interaction. The form of a message, whether it is presented as a question or a statement, has the potential to moderate the impact of the pick-up line type, because questions guide conversations by prompting a response (Dillon, 1982). Besides, it mitigates some negative perceptions associated with different types of pick-up lines, as the willingness to initiate and engage in conversations enhances the sender's perceived sociability (Dillon, 1982). However, no previous research has been conducted on the combination of type and form. Thus, the following sub-question is proposed:

Sub-question 1: "To what extent does the form of the pick-up line moderate the effect of the type on (a) perceived sociability and (b) the intention to respond?"

Gender differences

Although humor is an effective strategy to attract the attention of the opposite sex (Cunningham & Barbee, 2008; Senko & Fyffe, 2010), the way humor is perceived in online dating differs between individuals. Existing empirical results about gender differences are mixed, but previous research by Verwers (2021) has shown that humorous, or flippant, pick-up lines are perceived more positively among men than women. Men tend to favor flippant lines due to their playful and humorous tone, contributing to a relaxed vibe (Kleinke et al., 1986; Wade et al., 2009). The less serious nature of flippant lines seems to have less impact on men than on women, who are more likely to associate flippant lines with undesirable traits such as selfishness, dominance, and promiscuity (Kleinke & Dean, 1990). Women exhibit a strong preference for innocuous pick-up lines, as these are perceived as friendly, non-intrusive, and provide a safe way to initiate contact without suggesting sexual or romantic intent (Kleinke et al., 1986). Other studies suggest that both men and women tend to view direct complimentary lines as the most effective pick-up lines (Kleinke et al., 1986; Wade et al., 2009). These lines clearly convey intentions and are often associated with positive qualities, such as likability and responsibility (Kleinke & Dean, 1990). While some research indicates that women favor innocuous lines (Kleinke et al., 1986), others find no difference in women's evaluation of innocuous versus direct complimentary lines (Cunningham, 1989).

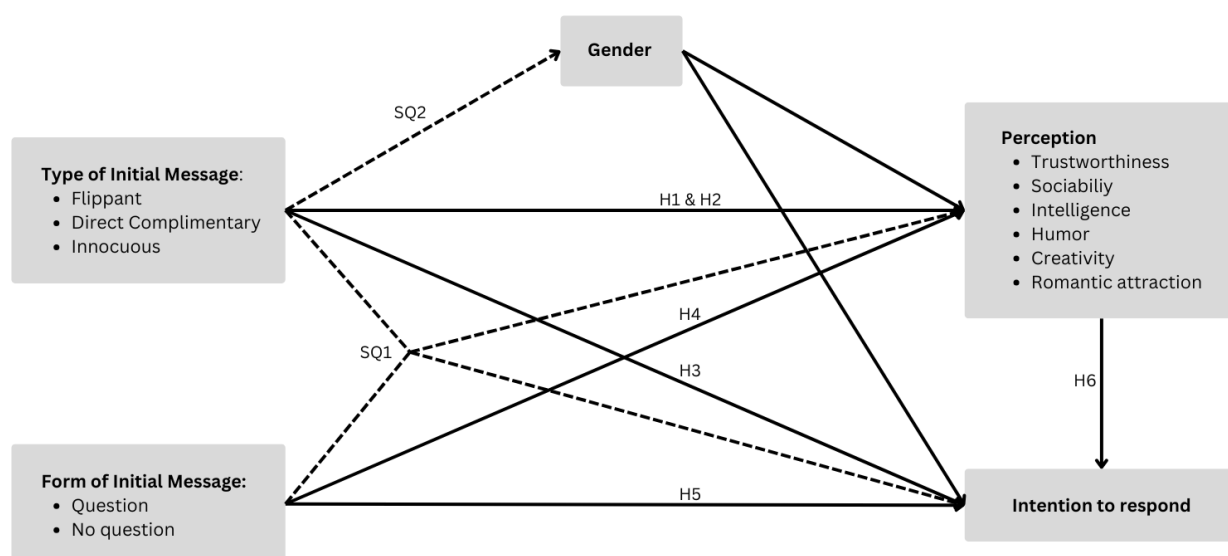
Given the varying research outcomes on gender differences, it is plausible that men and women may respond differently to flippant, direct complimentary, and innocuous pick-up lines. Therefore, it is essential to explore these gender-based differences to better understand their impact on online communication dynamics. The next sub-question is created:

Sub-question 2: “To what extent do gender differences moderate the effect of pick-up line type on the perception of the sender’s (a) humor, (b) creativity, (c) sociability, (d) romantic attraction, (e) trustworthiness, and (f) intelligence and on (g) the intention to respond?”

To better understand these relationships, Figure 1 illustrates the conceptual framework that guides the study.

Figure 1

Conceptual framework.



Method

Design

To investigate how different types of pick-up lines and message form influence the perception of the sender and intention to respond, a 3 (type: flippant, direct complimentary, and innocuous) $\times$ 2 (form: question versus no question) within-subjects design was employed.

Participants

Participants were recruited using a convenience sampling method, further supported by snowball sampling and social media to ensure diversity. Inclusion criteria required participants to be between 18 and 34 years old and to have used Tinder within the past six months. Participation was voluntary, and no compensation was provided.

Sixty Dutch individuals aged 18 to 34 were recruited for the study, as this demographic represents the most frequent users of dating apps like Tinder (Iqbal, 2024). However, three participants were excluded due to not meeting the recent Tinder activity criterion, resulting in a final sample of 57 participants. Among those, 30 identified as women and 27 as men. The average age was 23 years ($M = 23.4$, $SD = 2.40$).

Pre-test

Prior to the experiment, a pretest (Appendix A) was conducted with 18 potential pick-up lines to assess their appropriateness to represent each pick-up line type (flippant, direct complimentary, and innocuous) in a question and a statement. The sample consisted of 23 individuals, including 13 women and 10 men, with an average age of 25 years ($M = 25.0$, $SD = 4.14$). Participants rated 18 potential pick-up lines on a 7-point Likert scale (1 = "strongly disagree," 7 = "strongly agree") regarding their perceived humor, flirtatiousness, directness, complimentary nature, and underlying intentions. The pick-up lines were carefully constructed to align with Kleinke et al.'s (1986) criteria: flippant lines were expected to score high on perceived humor and flirtatiousness, direct complimentary lines high on perceived directness and complimentary nature, and innocuous lines low on directness and underlying intentions. By calculating the scores of the associated perceptions for each pick-up line and form, the six most appropriate ones were chosen to incorporate into the experiment (Appendix B). For the flippant pick-up line type, the selected lines were: "*Do you believe in love at first swipe or should I unmatched and try again?*" as a question (total score of 5.72), and "*I wasn't planning on falling in love here, but you're changing things.*" as a statement (total score of 5.60). The selected direct complimentary lines were: "*I like your smile, what makes you happy?*" as a question (total score of 5.46), and "*Hey, you have a nice vibe!*" as a statement (total score of 5.84). Lastly, for the innocuous pick-up line type, the selected lines were: "*Hey, do you know any good*

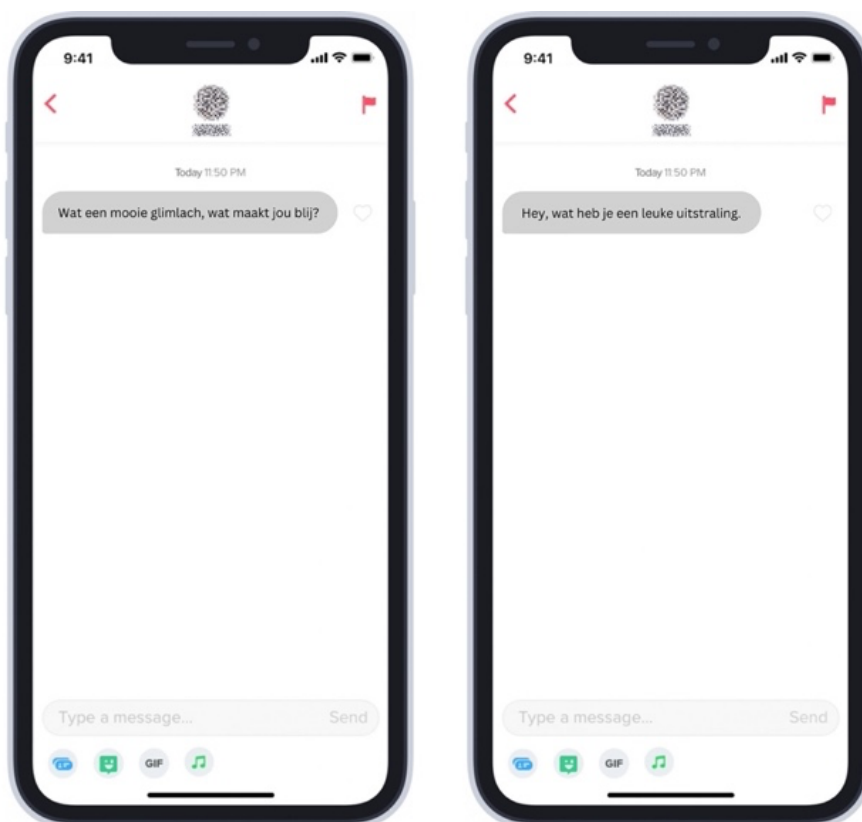
series?" as a question (total score of 2.53), and *"Hey, I'm also into traveling."* as a statement (total score of 2.62).

Materials

The stimuli (Appendix C) consisted of the six selected pick-up lines from the pre-test, representing three types of pick-up lines (flippant, direct complimentary, and innocuous) in both question and statement forms. To prevent bias and confounding variables based on the name or appearance of the sender, the messages were displayed in screenshots of Tinder chats with blurred usernames and profile pictures. This allowed participants to focus on the message content only. Participants were able to view the messages but could not type or respond to them. Each pick-up line type was presented in both a question-based (Figure 2a) and a statement-based (Figure 2b) form. Participants viewed all six conditions in a fixed order to maintain consistency across participants and reduce potential variations in responses caused by differences in sequence.

Figure 2

Examples of Tinder chats with (a) a direct complimentary question and (b) a direct complimentary statement.



Measurements

This study evaluated participants' perceptions of the sender and their intention to respond. All questions related to a specific pick-up line were presented together on a single page to ensure a focused assessment process. Adaptations were made to some existing scales to align with the context of online dating. Participants rated the perceptions of the sender based on desirable traits in a potential partner; humor, creativity, sociability, romantic attraction, trustworthiness, and intelligence (Senko & Fyffe, 2010). This included the following dimensions, each rated on a 7-point Likert scale (1 = "*strongly disagree*," 7 = "*strongly agree*"). Adapted from Brand et al. (2012), perceived humor was assessed using two items: "*I think this person has humor*" and "*I think this person is funny*" ($\alpha = .88$). Perceived creativity was measured based on two items of the Creative Personality Scale by Gough (1979): "*I think this person is creative*" and "*I think this person is original*" ($\alpha = .81$). Sociability was measured based on three items from Senko and Fyffe (2010): "*I think this person is sociable*", "*I think this person is friendly*", and "*I think this person is confident*" ($\alpha = .87$). Romantic attraction was measured using three items from Campbell's (1999) romantic attraction measure: "*I see this person as a desirable dating partner*", "*I would think positively about myself if I were dating this person*", and "*I think my friends would have a positive opinion about me if I were dating this person*" ($\alpha = .89$). Trustworthiness was measured using three items adapted from Senko and Fyffe (2010) "*I think this person is trustworthy*", "*I think this person is honest*", and "*I think this person is reliable*" ($\alpha = .93$). Finally, intelligence was assessed with two items adapted from Senko and Fyffe (2010): "*I think this person is smart*" and "*I think this person is intelligent*" ($\alpha = .92$).

Intention to respond was measured using a modified version of Warshaw and Davis' (1985) behavioral expectation scale, which refers to an individual's perceived probability of engaging in a specific future behavior. The scale was adapted to align with the current research and measured on a 7-point Likert scale (1 = "*strongly disagree*," 7 = "*strongly agree*") with three items: "*It is likely that I would reply to this person's message*", "*I would be excited to respond to this person*", and "*I would look forward to interacting with this person*" ($\alpha = .85$).

Procedure

Participants completed the survey (Appendix D) online via Qualtrics, providing convenience without time or location constraints (Treadwell, 2011). First, participants were informed about the research topic and their task. They were assured that participation was voluntary, confidential, and anonymous, and that they could withdraw at any time without a penalty or having their responses included in the analysis. After providing consent, participants answered screening questions to determine their recent activity on Tinder. Those who confirmed recent activity in the last six months proceeded, while others were excluded from the study. Participants were then asked to provide

demographic information, including age and gender. Next, they were presented with six pick-up lines, one at a time, and instructed to imagine receiving each message from a potential Tinder match. For each pick-up line, they responded to seven sets of questions designed to evaluate their perceptions and intention to respond. The survey finished with a thank-you message and the researcher's contact information for any follow-up questions.

Data Analysis

Data were analyzed using Jamovi. Two-way repeated measures ANOVAs examined the effects of pick-up line type and form on participants' perceptions of the sender and the intention to respond. This analysis examined both main effects and their interaction. Mauchly's test of sphericity was used to check assumptions, and Greenhouse-Geisser corrections were applied where sphericity was violated. Significant main effects were further analyzed with pairwise comparisons using paired-samples t-tests. Significant interactions were explored using simple effects analyses, with one-way repeated measures ANOVAs conducted for each message form, followed by paired-samples t-tests for pick-up line comparisons. Mediation analyses were conducted for each outcome variable using MedMod in Jamovi and gender differences were explored as a potential moderator through additional two-way repeated measures ANOVAs, with post hoc comparisons to further interpret interactions.

Results

To investigate how different types of pick-up lines (flippant, direct complimentary, and innocuous) and message form (question versus no question) impact perceptions of the sender and the intention to respond on online dating apps, a series of two-way repeated measures ANOVAs were performed, using Jamovi. Although we only have an expectation regarding the effect of the form of a pick-up line on sociability, we will examine all perceptions, as these have not been previously explored in research. Post hoc comparisons were performed to interpret significant effects and the moderating effects of perceptions and gender were examined. Table 1 presents the means and standard deviations for perceived humor, creativity, sociability, romantic attraction, trustworthiness, intelligence, and intention to respond to provide an overview of how each pick-up line type was evaluated.

Table 1

Means and standard deviations for perceived humor, creativity, sociability, romantic attraction, trustworthiness, intelligence, and intention to respond across the three types of pick-up lines, with FL standing for Flippant, DC for Direct complimentary and IN for Innocuous.

	Humor	Creativity	Sociability	Romantic attraction	Trustworthiness	Intelligence	Intention to respond
FL	3.96 (1.41) ^a	3.59 (1.29) ^a	4.59(1.09) ^a	3.21 (1.27) ^a	3.02 (1.10) ^a	3.59 (1.11) ^a	3.68 (1.50) ^a
DC	3.16 (1.04) ^b	3.25(1.13) ^a	4.61(0.94) ^b	3.86 (1.35) ^a	4.24 (1.11) ^b	4.04 (1.17) ^b	4.29 (1.44) ^b
IN	2.45 (1.16) ^c	2.49 (1.39) ^b	3.26(1.08) ^c	2.69 (1.24) ^b	3.89 (1.25) ^c	3.09 (1.29) ^c	2.74 (1.36) ^c

Note: Superscripts (^a, ^b, ^c) indicate significant differences within each column. Means with the same superscript do not differ significantly; different superscripts indicate a significant difference.

Humor

In the case of humor, both the assumptions of normality and sphericity ($p = .210$) were met. A significant main effect of pick-up line type on humor was found, $F(2,112) = 33.10$, $p < .001$, with a large effect size of $\eta^2 = .17$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the flippant pick-up lines ($M = 3.96$, $SD = 1.41$) as significantly more humoristic than the direct complimentary lines ($M = 3.16$, $SD = 1.04$), $p < .001$, $d = -0.57$, 95% CI [-0.85, -0.29], and the innocuous lines ($M = 2.45$, $SD = 1.16$), $p < .001$, $d = 0.90$, 95% CI [0.59, 1.21]. Besides, direct complimentary lines were perceived significantly higher in humor compared to innocuous pick-up lines, $p < .001$, $d = 0.68$, 95% CI [0.39, 0.96].

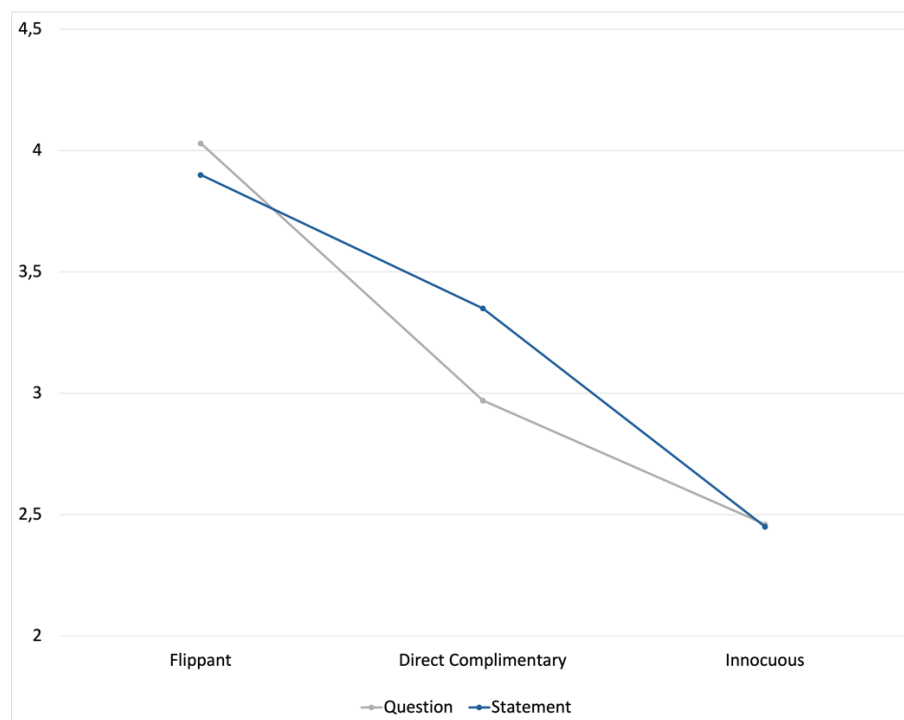
There was no significant main effect of form on humor, $F(1,56) = 0.66$, $p = .419$, $\eta^2 = .001$, but the interaction between type and form was significant, $F(2,112) = 3.10$, $p = .049$, with a small effect

size of $\eta^2 = .01$ (Figure 3). The simple effects of pick-up line types were explored in light of the significant type x form interaction. When the pick-up lines were direct complimentary, the simple effect of form was significant, $F(1, 56) = 6.17, p = .016, \eta^2 = .03$, indicating that the form (question vs. statement) influenced perceived humor. However, no significant simple effects of form were found for the flippant lines, $F(1, 56) = 0.43, p = .515, \eta^2 = .00$ and the innocuous lines, $F(1, 56) = 0.00, p = .946, \eta^2 = .00$. Unadjusted post hoc simple comparisons revealed that direct complimentary questions ($M = 2.97, SD = 1.26$), were rated significantly lower in perceived humor than direct complimentary statements ($M = 3.35, SD = 1.22$), $p = .016, d = -0.33, 95\% CI [-0.59, -0.06]$.

These findings support H1a, indicating flippant lines were perceived as more humoristic than the other types.

Figure 3

The interaction effect of pick-up line type and form on perceived humor.



Creativity

Regarding creativity, the assumptions of normality and sphericity ($p = .266$) were both met. A significant main effect of pick-up line type on creativity was found, $F(2,112) = 13.30, p < .001$, with a medium effect size of $\eta^2 = .10$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the flippant pick-up lines ($M = 3.59, SD = 1.29$) as significantly more creative than the innocuous line ($M = 2.49, SD = 1.39$), $p < .001, d = 0.60, 95\% CI [0.31, 0.88]$, but there was no significant difference between the flippant and direct complimentary pick-up lines ($M = 3.25, SD =$

1.13), $p = .113$, $d = -0.21$, 95% CI [-0.48, 0.05]. Participants rated the innocuous pick-up lines as significantly less creative than the direct complimentary lines, $p < .001$, $d = 0.76$, 95% CI [0.24, 0.79].

However, there was no significant main effect of form on creativity, $F(1,56) = 0.08$, $p = .778$, $\eta^2 = .000$, and the interaction between type and form was not significant, $F(2,112) = 1.90$, $p = .155$, $\eta^2 = .004$.

These results do not support H1b, as there was no significant difference between the flippant and direct complimentary lines.

Sociability

As for sociability, the assumptions of normality and sphericity ($p = .603$) were both met. A significant main effect of pick-up line type on sociability was found, $F(2,112) = 48.05$, $p < .001$, with a large effect size of $\eta^2 = .22$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the innocuous pick-up lines ($M = 3.26$, $SD = 1.08$) as significantly less sociable than the direct complimentary lines ($M = 4.61$, $SD = 0.94$), $p < .001$, $d = -1.20$, 95% CI [0.86, 1.54], and the flippant lines ($M = 4.59$, $SD = 1.09$), $p < .001$, $d = 0.94$, 95% CI [0.62, 1.25]. However, there was no significant difference between the direct complimentary and flippant lines, $p = .894$, $d = 0.02$, 95% CI [-0.24, 0.28].

Moreover, there was no significant main effect of form on sociability, $F(1,56) = 0.13$, $p = .724$, $\eta^2 = .00$, and the interaction between type and form was not significant, $F(2,112) = 0.47$, $p = .626$, $\eta^2 = .00$.

These results do not support H1c, as flippant pick-up lines were perceived more sociable than innocuous lines, but there was no significant difference with the direct complimentary lines. They do align with H2a, because direct complimentary pick-up lines were perceived more sociable than innocuous lines. However, these findings do not support H4, as no significant difference was found between pick-up lines phrased as questions or statements. Additionally, these results answered sub-question 1a, as the form of the message did not moderate the effect of message type on perceived sociability.

Romantic attraction

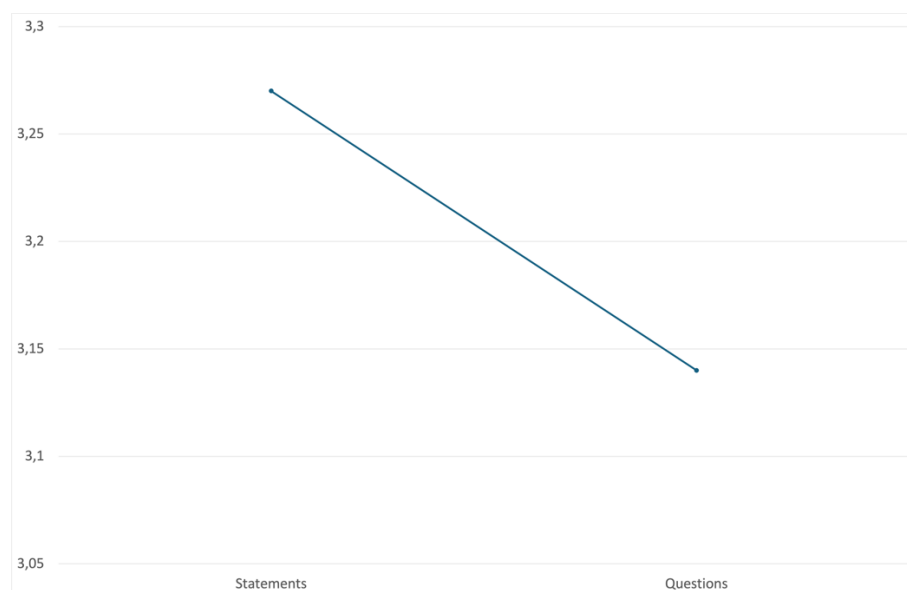
Regarding romantic attraction, the assumptions of normality and sphericity ($p = .149$) were both met. A significant main effect of pick-up line type on romantic attraction was found, $F(2,112) = 15.15$, $p < .001$, with a medium effect size of $\eta^2 = .10$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the direct complimentary pick-up lines ($M = 3.86$, $SD = 1.35$) as significantly more likely to elicit romantic attraction than the flippant lines ($M = 3.21$, $SD = 1.27$), $p = .003$, $d = 0.41$, 95% CI [0.14, 0.68] and the innocuous lines ($M = 2.69$, $SD = 1.24$), $p < .001$, $d = 0.72$,

95% CI [0.42, 1.01]. Additionally, participants rated the flippant pick-up lines as significantly more likely to elicit romantic attraction than the innocuous lines, $p = .017$, $d = 0.33$, 95% CI [0.06, 0.59].

Besides, there was a significant main effect of form on romantic attraction, $F(1,56) = 6.80$, $p = .012$, $\eta^2 = .01$ (Figure 4). Unadjusted post hoc comparisons revealed that pick-up lines phrased as statements ($M = 3.27$, $SD = 1.00$), were rated significantly higher on romantic attraction than pick-up lines phrased as questions ($M = 3.14$, $SD = 0.92$), $p = .012$, $d = -0.35$, 95% CI [-0.61, -0.08].

Figure 4

The main effect of message form on perceived romantic interest.



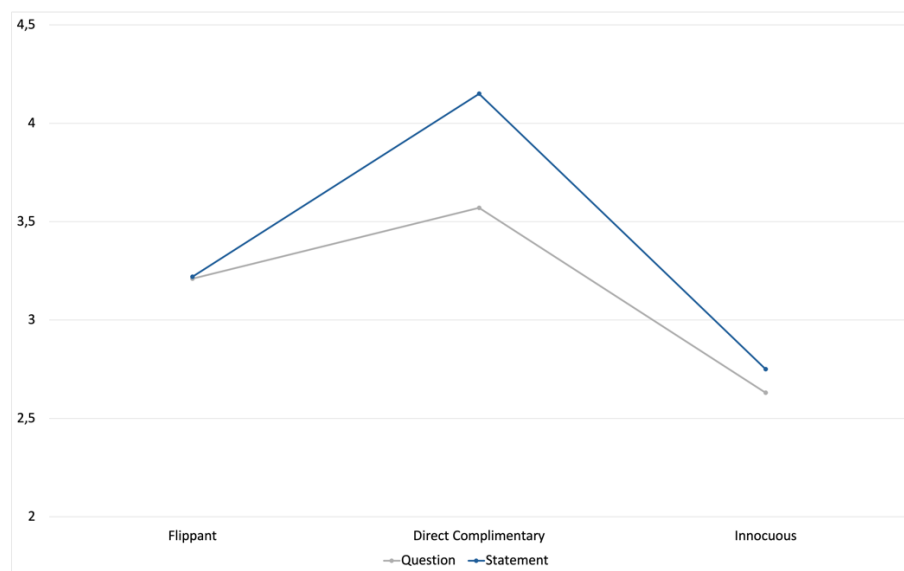
Moreover, there was a significant interaction between pick-up line type and form, $F(2,112) = 3.69$, $p = .028$, with a small effect size of $\eta^2 = .01$ (Figure 5). The simple effects of pick-up line types were explored in light of the significant type x form interaction. When pick-up lines were phrased as questions, the simple effect of pick-up line type was significant, $F(2, 112) = 7.43$, $p < .001$, $\eta^2 = .071$. Similarly, when pick-up lines were phrased as statements, the effect remained significant, $F(2, 112) = 18.2$, $p < .001$, $\eta^2 = .143$, warranting examination of simple comparisons within each form. Unadjusted post hoc simple comparisons revealed that innocuous questions ($M = 2.63$, $SD = 1.30$), were rated significantly lower in eliciting romantic attraction than both direct complimentary questions ($M = 3.57$, $SD = 1.50$), $p < .001$, $d = 0.49$, 95% CI [0.22, 0.77] and flippant questions ($M = 3.21$, $SD = 1.41$), $p = .024$, $d = -0.31$, 95% CI [-0.57, -0.04]. However, there was no significant difference between direct complimentary questions and flippant questions, $p = .133$, $d = 0.20$, 95% CI [-0.06, 0.46]. Similarly, post hoc simple comparisons revealed that innocuous statements ($M = 2.75$, $SD = 1.39$), were again rated significantly less likely to elicit romantic attraction than both direct

complimentary statements ($M = 4.15$, $SD = 1.55$), $p < .001$, $d = -0.74$, 95% CI [-1.03, -0.45] and flippant statements ($M = 3.22$, $SD = 1.35$), $p = .043$, $d = 0.27$, 95% CI [0.01, 0.54]. Additionally, flippant statements were significantly less likely to elicit romantic attraction than direct complimentary statements, $p < .001$, $d = 0.53$, 95% CI [0.25, 0.81].

These findings do not align with H1d, as flippant pick-up lines elicit more romantic attraction than innocuous lines, but less compared to direct complimentary lines. However, the findings do support H2b, indicating direct complimentary pick-up lines elicit more romantic attraction than innocuous lines.

Figure 5

The interaction effect of pick-up line type and form on perceived romantic interest.



Trustworthiness

Regarding trustworthiness, Mauchly's test of sphericity indicated a sphericity breach for the interaction test ($p = .039$). Consequently, Greenhouse-Geisser epsilon corrected results are reported for all omnibus main effects, interactions, and simple effects. The assumption of normality was met. A significant main effect of pick-up line type on trustworthiness was found, $\epsilon = .86$, $F(1.71, 96.04) = 31.95$, $p < .001$, with a large effect size of $\eta^2 = .13$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the trustworthiness of the flippant pick-up lines ($M = 3.02$, $SD = 1.10$) significantly lower than the direct complimentary pick-up lines ($M = 4.24$, $SD = 1.11$), $p < .001$, $d = 1.08$, 95% CI [0.75, 1.40] and the innocuous pick-up lines ($M = 3.89$, $SD = 1.25$), $p < .001$, $d = -0.62$, 95% CI [-0.90, -0.34]. Besides, the innocuous lines were rated significantly less trustworthy than the direct complimentary pick-up lines, $p = .010$, $d = 0.35$, 95% CI [0.08, 0.62].

In contrast, there was no significant main effect of form on trustworthiness, $F(0.38, 28.18) = 0.75$, $p = .391$, $\eta^2 = .00$. Similarly, the interaction between type and form was not significant, $F(1.11, 112.78) = 0.55$, $p = .560$, $\eta^2 = .00$.

These results support H1e, indicating flippant lines were perceived as less trustworthy than the other types.

Intelligence

For intelligence, the assumptions of normality and sphericity ($p = .272$) were both met. A significant main effect of pick-up line type on intelligence was found, $F(2, 112) = 19.74$, $p < .001$, with a medium effect size of $\eta^2 = .09$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the flippant pick-up lines ($M = 3.59$, $SD = 1.11$) as significantly less intelligent than the direct complimentary lines ($M = 4.04$, $SD = 1.17$), $p < .001$, $d = 0.59$, 95% CI [0.31, 0.87], but significantly more intelligent than innocuous lines ($M = 3.09$, $SD = 1.29$), $p = .004$, $d = 0.40$, 95% CI [0.13, 0.67]. Additionally, direct complimentary lines were rated as significantly more intelligent than the innocuous lines, $p < .001$, $d = 0.75$, 95% CI [0.45, 1.04].

However, there was no significant main effect of form on intelligence, $F(1, 56) = 0.34$, $p = .561$, $\eta^2 = .00$, and the interaction between type and form was not significant, $F(2, 112) = 2.91$, $p = .059$, $\eta^2 = .01$.

These results do not align with H1f, because flippant lines were perceived as less intelligent than direct complimentary lines, but more intelligent compared to innocuous lines.

Intention to respond

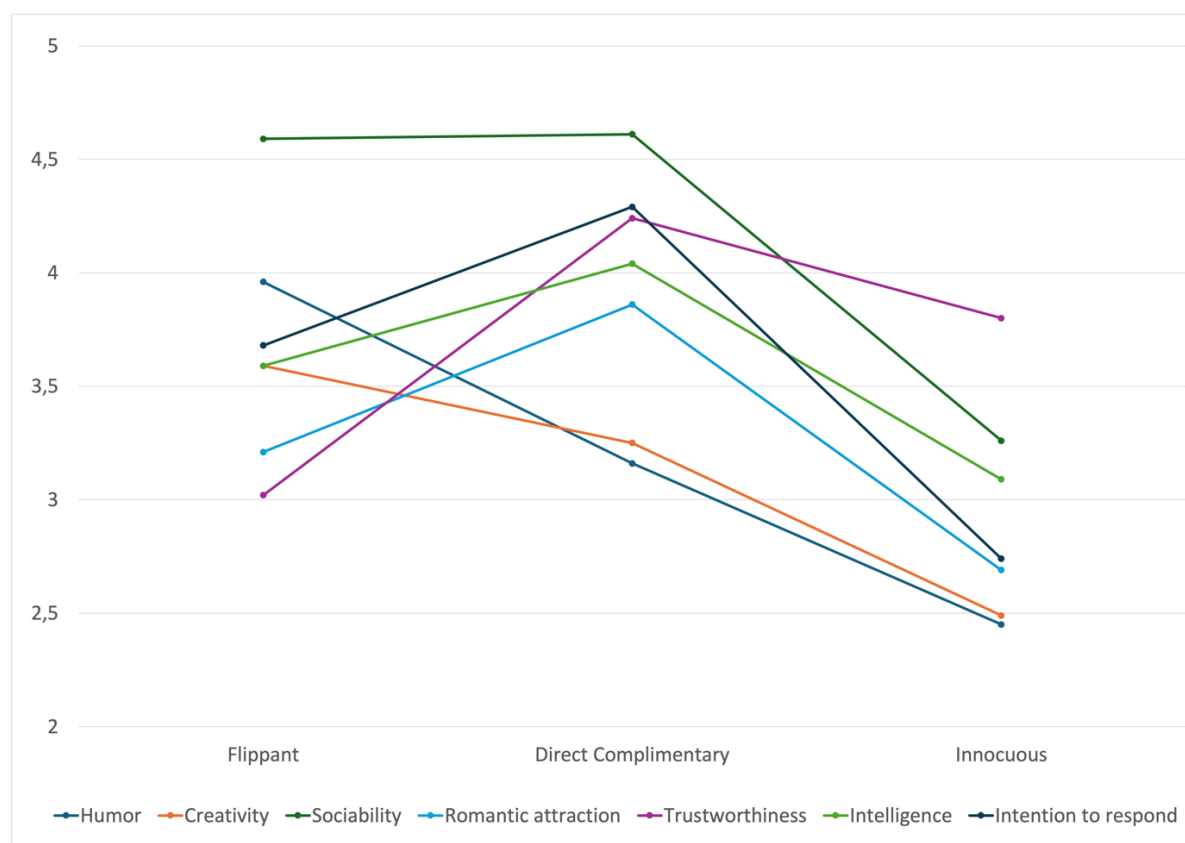
When assessing intention to respond, both the assumptions of normality and sphericity ($p = .874$) were met. The analysis revealed a significant main effect of pick-up line type on intention to respond, $F(2, 112) = 18.92$, $p < .001$, with a large effect size of $\eta^2 = .13$ (Figure 6). Unadjusted post hoc comparisons revealed that participants rated the direct complimentary pick-up lines ($M = 4.29$, $SD = 1.44$) as significantly more likely to elicit an intention to respond than the flippant lines ($M = 3.68$, $SD = 1.50$), $p = .012$, $d = -0.34$, 95% CI [-0.61, -0.08] and the innocuous lines ($M = 2.74$, $SD = 1.36$), $p < .001$, $d = 0.81$, 95% CI [0.50, 1.10]. Additionally, participants rated the flippant pick-up lines as significantly more likely to elicit romantic attraction than the innocuous lines, $p < .001$, $d = 0.46$, 95% CI [0.19, 0.73].

However, there was no significant main effect of form on intention to respond, $F(1, 56) = 3.82$, $p = .055$, $\eta^2 = .01$, and the interaction between type and form was not significant, $F(2, 112) = 1.84$, $p = .163$, $\eta^2 = .00$.

These results do not support H3, because flippant pick-up lines resulted in lower intentions to respond compared to direct complimentary, but higher intentions to respond compared to innocuous lines. Besides, direct complimentary pick-up lines were rated higher on intention to respond than innocuous lines. These findings also do not align with H5, as no significant difference was found between pick-up lines phrased as questions or statements. Moreover, the results answered sub-question 1b, as the form of the message did not moderate the effect of message type on intentions to respond.

Figure 6

The main effect of pick-up line type on perceived humor, creativity, sociability, romantic attraction, trustworthiness, intelligence, and intention to respond.



Mediation analyses

To test whether perceived humor, creativity, sociability, romantic attraction, trustworthiness, and intelligence mediate the relationship between the effect of message type on the intention to respond, mediation analyses were done in Jamovi.

For humor, the results showed a significant total effect of pick-up line type on the intention to respond, $b = -0.62$, $SE = 0.16$, $z = -3.97$, $p < .001$, as well as a significant direct effect, $b = -0.43$, $SE = 0.12$, $z = -3.59$, $p < .001$. However, the indirect effect of humor as a mediator was not significant, b

= -0.19, $SE = 0.10$, $z = -1.85$, $p = .064$. While humor significantly influenced the intention to respond, $b = 0.74$, $SE = 0.07$, $z = 11.27$, $p < .001$, the type of pick-up line did not significantly affect humor, $b = -0.26$, $SE = 0.14$, $z = -1.88$, $p = .061$. These findings do not support H6a by suggesting that perceived humor does not significantly mediate the relationship between pick-up line type and the intention to respond. In contrast, the effect of pick-up line type on the intention to respond appears to be direct.

Creativity revealed a significant total effect of pick-up line type on the intention to respond, $b = -0.62$, $SE = 0.16$, $z = -3.97$, $p < .001$, with both a significant direct effect, $b = -0.25$, $SE = 0.11$, $z = -2.15$, $p = .032$, and a significant indirect effect with creativity as a mediator, $b = -0.37$, $SE = 0.11$, $z = -3.29$, $p = .001$. The type of pick-up line affects perceived creativity, $b = -0.44$, $SE = 0.13$, $z = -3.40$, $p < .001$, and creativity significantly influences the intention to respond, $b = 0.85$, $SE = 0.07$, $z = 12.99$, $p < .001$. These results show that perceived creativity partially mediates the relationship between pick-up line type and the intention to respond. As the mediation is partial, H6b is not supported.

In the case of sociability, the total effect of pick-up line type on intention to respond was significant, $b = -0.62$, $SE = 0.16$, $z = -3.97$, $p < .001$. However, the direct effect of pick-up line type on intention to respond was not significant, $b = 0.01$, $SE = 0.13$, $z = 0.11$, $p = .916$, while the indirect effect with sociability as a mediator was significant, $b = -0.63$, $SE = 0.12$, $z = -5.43$, $p < .001$. The type of pick-up line significantly influenced sociability, $b = -0.68$, $SE = 0.11$, $z = -6.17$, $p < .001$, and sociability positively significantly influences the intention to respond, $b = 0.93$, $SE = 0.08$, $z = 11.41$, $p < .001$. These findings show that perceived sociability fully mediates the relationship between pick-up line type and the intention to respond, as the direct effect of type on response intention is no longer significant when sociability is considered (Figure 13, Appendix E). Therefore, H6c is supported.

Romantic attraction emerged as a significant mediator, with a total effect of pick-up line type on intention to respond, $b = -0.62$, $SE = 0.16$, $z = -3.97$, $p < .001$. However, the direct effect of pick-up line type on intention to respond was not significant, $b = -0.15$, $SE = 0.09$, $z = -1.67$, $p = .095$, while the indirect effect with romantic attraction as a mediator was significant, $b = -0.47$, $SE = 0.13$, $z = -3.57$, $p < .001$. The type of pick-up line significantly influenced romantic attraction, $b = -0.47$, $SE = 0.13$, $z = -3.63$, $p < .001$, and romantic attraction significantly positively influences the intention to respond, $b = 0.99$, $SE = 0.05$, $z = 18.82$, $p < .001$. These findings support H6d by showing a full mediation of perceived romantic attraction on the relationship between pick-up line type and the intention to respond, as the direct effect of type on intention to respond is no longer significant when romantic attraction is considered (Figure 14, Appendix E).

For trustworthiness, the results showed a significant total effect of pick-up line type on the intention to respond, $b = -0.62$, $SE = 0.16$, $z = -3.97$, $p < .001$, as well as a significant direct effect, $b = -0.56$, $SE = 0.14$, $z = -3.95$, $p < .001$. However, the indirect effect of trustworthiness as a mediator was not significant, $b = -0.06$, $SE = 0.07$, $z = -0.88$, $p = .373$. While trustworthiness significantly

influenced the intention to respond, $b = 0.52$, $SE = 0.09$, $z = 5.94$, $p < .001$, the type of pick-up line did not significantly affect trustworthiness $b = -0.11$, $SE = 0.12$, $z = -0.89$, $p = .373$. This does not support H6e by indicating that trustworthiness does not mediate the relationship between pick-up line type and the intention to respond. Instead, the effect of pick-up line type on intention to respond is direct.

Finally, for intelligence, the results showed a significant total effect of pick-up line type on the intention to respond, $b = -0.62$, $SE = 0.16$, $z = -3.97$, $p < .001$, with both a significant direct effect, $b = -0.30$, $SE = 0.13$, $z = -2.30$, $p = .022$, and a significant indirect effect with intelligence as a mediator, $b = -0.32$, $SE = 0.10$, $z = -3.28$, $p = .001$. The type of pick-up line significantly affects perceived intelligence, $b = -0.42$, $SE = 0.12$, $z = -3.50$, $p < .001$, and intelligence significantly influences the intention to respond, $b = 0.76$, $SE = 0.08$, $z = 9.42$, $p < .001$. These results shows that perceived intelligence partially mediates the relationship between pick-up line type and the intention to respond. As the mediation is partial, H6f is not supported.

Gender differences

Additional analyses were done to investigate how gender differences moderate the effect of message type on perception.

A two-way repeated measures ANOVA was conducted to examine whether gender moderates the effect of pick-up line type on humor. Mauchly's test of sphericity indicated a sphericity breach for the interaction test ($p < .001$) and so, for consistency, Greenhouse-Geisser epsilon corrected results are reported for all omnibus main effects, interactions, and simple effects. The assumption of normality was met. No significant moderating effect of gender was found, $\varepsilon = .798$, $F(1.60, 87.80) = 0.26$, $p = .720$, $\eta^2 = .00$.

A two-way repeated measures ANOVA also assessed whether gender moderates the effect of pick-up line type on creativity. Both the assumptions of normality and sphericity ($p = .128$) were met. No significant moderating effect of gender was found, $F(2,110) = 0.11$, $p = .899$, $\eta^2 = .00$.

Another two-way repeated measures ANOVA was conducted to examine whether gender moderates the effect of pick-up line type on sociability. Mauchly's test of sphericity indicated a sphericity breach for the interaction test ($p = .004$) and so, for consistency, Greenhouse-Geisser epsilon corrected results are reported for all omnibus main effects, interactions, and simple effects. The assumption of normality was met. No significant moderating effect of gender was found, $\varepsilon = .844$, $F(1.69, 92.82) = 0.72$, $p = .468$, $\eta^2 = .00$.

For romantic attraction, a two-way repeated measures ANOVA revealed that both the assumptions of normality and sphericity ($p = .406$) were met. A significant main effect of gender was found, $F(2,110) = 7.86$, $p < .001$, $\eta^2 = .06$. Unadjusted post hoc simple comparisons revealed that women ($M = 3.98$, $SD = 1.09$) rated the flippant pick-up lines significantly lower in romantic attraction

than men ($M = 2.52$, $SD = 1.01$), $p < .001$, $d = 1.39$, 95% CI [0.81, 1.97]. However, there was no significant difference for the direct complimentary lines when rated by men ($M = 3.80$, $SD = 0.99$), or women ($M = 3.92$, $SD = 1.62$), $p = .740$, $d = -0.09$, 95% CI [-0.61, 0.43]. Besides, there was no significant difference for the innocuous lines when rated by men ($M = 2.96$, $SD = 1.09$), or women ($M = 2.46$), $SD = 1.33$, $p = .325$, $d = 0.41$, 95% CI [-0.12, 0.91].

To investigate how gender differences moderate the effect of pick-up line type on trustworthiness, another two-way repeated measures ANOVA was conducted. Mauchly's test of sphericity indicated a sphericity breach for the interaction test ($p = .016$) and so, for consistency, Greenhouse-Geisser epsilon corrected results are reported for all omnibus main effects, interactions, and simple effects. The assumption of normality was met. A significant moderating effect of gender was found, $\epsilon = .876$, $F(1.75, 96.39) = 10.2$, $p < .001$, $\eta^2 = .05$. Unadjusted post hoc simple comparisons revealed that women ($M = 2.56$, $SD = 0.93$) rated the flippant pick-up lines significantly lower in trustworthiness than men ($M = 3.52$, $SD = 1.06$), $p < .001$, $d = 0.97$, 95% CI [0.42, 1.52]. However, there was no significant difference for the direct complimentary lines when rated by men ($M = 4.14$, $SD = 0.87$), or women ($M = 4.33$, $SD = 1.30$), $p = .508$, $d = -0.18$, 95% CI [-0.70, 0.35]. Besides, there was no significant difference for the innocuous lines when rated by men ($M = 3.80$, $SD = 1.13$), or women ($M = 3.97$, $SD = 1.37$), $p = .626$, $d = -0.13$, 95% CI [-0.65, 0.39].

A two-way repeated measures ANOVA also assessed whether gender moderates the effect of pick-up line type on intelligence. Mauchly's test of sphericity indicated a sphericity breach for the interaction test ($p < .001$) and so, for consistency, Greenhouse-Geisser epsilon corrected results are reported for all omnibus main effects, interactions, and simple effects. The assumption of normality was met. No significant moderating effect of gender was found, $\epsilon = .761$, $F(1.52, 83.67) = 0.65$, $p = .483$, $\eta^2 = .00$.

Another two-way repeated measures ANOVA was conducted to examine whether gender moderates the effect of pick-up line type on the intention to respond. Both the assumptions of normality and sphericity ($p = .400$) were met. No significant moderating effect of gender was found, $F(2,110) = 2.35$, $p = .100$, $\eta^2 = .02$.

These results answered sub-question 2, by suggesting that gender differences significantly moderate the effect of message type on the perception of the sender's (e) trustworthiness and (d) romantic attraction, but not on the perceived (a) humor, (b) creativity, (c) sociability, (f) intelligence and on (g) the intention to respond.

Discussion

The present study aimed to investigate how different types of pick-up lines (flippant, direct complimentary, innocuous) and message forms (question versus no question) affect perceptions of the sender and the intention to respond among Dutch young adults on dating apps.

Hypotheses 1 and 2 focused on the impact of the type of pick-up line on the perception of humor, creativity, sociability, romantic attraction, trustworthiness, and intelligence, while hypothesis 3 explored the effect of the type on the intention to respond. Furthermore, hypotheses 4 and 5 investigated the influence of message form on perceptions and the intention to respond. Additionally, hypothesis 6 examined whether the effects of different types of pick-up lines on the intention to respond would be mediated by the perceptions of the receiver. Finally, sub-questions examined the interactions between message type and form, as well as potential gender differences. By investigating these effects, this study contributes to understanding how pick-up lines shape first impressions and highlights the complexities of interactions in online dating.

This research shows that different types of pick-up lines and message form influence the perception and intention to respond on online dating apps, with direct complimentary lines generally rated highest in sociability, romantic attraction, trustworthiness, intelligence, and intention to respond. However, the effect of message form is limited, except for its ability to enhance romantic attraction when phrased as statements.

Findings of types

The type of pick-up line plays a significant role in shaping perceptions. Flippant pick-up lines, often seen as humorous, were perceived lowest for trustworthiness, aligning with prior research highlighting the negative perceptions associated with flippant lines in offline settings (Levine et al., 1994; Kleinke & Dean, 1990). Despite this, flippant lines were perceived as the most humorous and creative compared to the other types. These findings confirm research emphasizing the role of humor in successful flirting in online dating (Apostolou & Christoforou, 2020). However, flippant lines elicited lower romantic attraction than direct complimentary lines, which are straightforward compliments. This suggests that while humor may enhance likeability, it might not necessarily translate to romantic attraction, perhaps due to perceived lack of seriousness (Cunningham, 1989). This could be explored in future research to better understand the relationship between humor and romantic attraction.

Direct complimentary pick-up lines emerged as most effective overall. They were rated highest for sociability, romantic attraction, trustworthiness, and intelligence, and were consistently assessed better than the more subtle and indirect innocuous lines. This aligns with findings that direct expressions of interest are associated with higher attractiveness (Kleinke & Dean, 1990) and

sociability (Senko & Fyffe, 2010). Since sociability is linked to confidence (Senko and Fyffe, 2010), the high rating on sociability suggests that direct complimentary lines are perceived as confident, conveying clear intent associated with self-assuredness in online dating (Kleinke & Dean, 1990). Innocuous lines were rated less positive across all dimensions likely due to their lack of engaging or creative appeal. According to Masden and Edwards (2015), originality is essential to create positive first impressions.

Previous studies on the use of pick-up lines in offline settings showed that individuals tend to favor innocuous lines, followed by direct complimentary lines, while flippant lines are considered less effective (Kleinke & Dean, 1990). However, this study, conducted nearly 35 years later in an online dating context, resulted in the most intention to respond with direct complimentary pick-up lines, followed by flippant lines, with the innocuous lines considered as least effective. This shift may reflect changes in social dynamics and communication styles over time, particularly with the rise of online interactions. Another possible explanation can be attributed to the abundance of choices on dating apps and the absence of non-verbal cues (Burgoon et al., 2021). In online dating settings, where individuals face numerous potential matches (Pronk & Denissen, 2020), direct complimentary lines may be more likely to capture attention because they clearly convey positive intentions (Doohan & Manusov, 2004). Besides, without non-verbal cues, the style of the pick-up line becomes more important in conveying intentions (Burgoon et al., 2021), making direct complimentary lines potentially more effective. Further research could explore whether emojis compensate for the lack of non-verbal cues in online communication. This includes analyzing whether the use of emojis can enhance the effectiveness of pick-up lines and if their impact varies depending on the type of pick-up line.

Impact of form

Surprisingly, message form (questions vs. statements) did not significantly influence most perceptions and the intention to respond. This is in contrast to research suggesting that questions increase reciprocity and engagement (Dillon, 1982), possibly due to the limited variation in pick-up lines that may have restricted the ability to observe a more detailed effect on perceptions and the intention to respond. Pick-up lines framed as statements were rated higher in romantic attraction than when they were framed as questions. This might indicate that the declarative nature of a statement may come across as more confident than a question, which could be perceived as less certain (Presley, 2022). Statements might appear more self-assured, aligning with traits that are associated with romantic interest (Buunk et al., 2002).

The link between type and form

This study also examined how pick-up line type and message form interact, influencing perceptions and the intention to respond. The results suggest that the outcomes are primarily influenced by the type of pick-up line, though the form also affects perceptions of humor and romantic attraction depending on the type of pick-up line. Specifically, pick-up lines framed as questions were only perceived less humorous than statements when it was a direct complimentary pick-up line. For romantic attraction, the same effect is the strongest for direct complimentary lines, while there was no interaction effect found for flippant lines. The preference for statements may be related to their perceived assertiveness and confidence (Presley, 2022). These findings highlight how specific combinations of pick-up line type and form influence perceptions, offering insights into interpersonal communication dynamics. Further research could employ qualitative interviews to explore how the interplay between message form and perceived traits such as confidence contributes to these outcomes.

The role of romantic attraction and sociability

The type of pick-up line influenced the intention to respond mainly through perceptions of romantic attraction and sociability. This is consistent with research from Meier et al. (2010), suggesting that sociability influences perceived attractiveness, and Rosenberg and Tunney (2008), who found that romantic attraction influences the likelihood of receiving a response. Creativity and intelligence played a smaller role, while humor and trustworthiness had no significant impact. Future research could explore how dating app users can craft pick-up lines to enhance perceptions of sociability, romantic attraction, creativity, and intelligence, and whether this increases the intention to respond.

Gender differences

Regarding the role of gender in online dating communication, direct compliments elicited less romantic attraction among women compared to men. Besides, trustworthiness was rated lower by women in response to direct complimentary lines, which is consistent with literature on women's preference for safer, non-intrusive approaches (Kleinke et al., 1986). No gender differences were found for humor, creativity, sociability, intelligence, and intention to respond. This is in contrast with prior research by Verwers (2021), indicating that humor in pick-up lines is perceived differently by men and women. These findings highlight that while perceptions of romantic attraction and trustworthiness differ between men and women, other qualities are perceived similarly by both genders in response to different types of pick-up lines. This identification of gender differences offers

new opportunities for future research on how individuals differ in their interactions and expectations in romantic communication.

Limitations

Although this study provides some interesting findings, several limitations should be acknowledged. Participants were presented Tinder chats with blurred usernames and profile pictures. While this approach ensured the focus remained on the message content, it reduced ecological validity as profile pictures and names may influence first impressions and responses on dating apps (Brand et al., 2012). Future research could include profile pictures and names to align with realistic dating profiles, allowing for an examination of how these features interact with message content to influence perceptions and responses. Such an approach may reveal that factors like perceived attractiveness and positive traits (e.g., friendliness) in a dating profile are significant predictors of dating intentions (Dai & Robbins, 2020).

Furthermore, the study used one line per condition. This limited variation in message stimuli may not fully capture the diversity of pick-up lines in real dating contexts. Each condition was represented by only one pick-up line with a difference in length and subject. Future studies should consider using multiple messages for each condition to capture a wider range of variations within each category.

Moreover, the within-subjects design of this study may have introduced carryover effects (Greenwald, 1976), where exposure to earlier conditions influenced participants' perceptions and responses to later conditions. While this reflects real dating app experiences, where users see multiple pick-up lines one after another, future studies could use a between-subjects design, with each participant evaluating one condition to address this potential confound.

By addressing these limitations, future studies can build on the findings of this study to better understand the complexities of interactions and user dynamics in online dating contexts. Follow-up research could involve participants typing responses to the pick-up lines to analyze how different message types influence actual engagement. This would provide further insights into how perceptions impact ongoing conversations in online dating.

Conclusion

This study emphasizes the effectiveness of direct complimentary pick-up lines, enhancing sociability, romantic attraction, trustworthiness, and intelligence in online dating settings, while innocuous or flippant approaches are less preferred. Dating apps can leverage these insights by giving users tips on writing effective first messages, to enhance the intention to respond and foster romantic connections. Additionally, developers of dating apps could develop algorithms to analyze

successful messaging patterns in different contexts to provide tailored communication strategies. These findings offer valuable insights for people looking for a potential partner online and highlight the importance of clear and confident communication in online dating settings, where non-verbal cues are absent.

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Appendix A

Pre-Test

Instructions

Bedankt voor je deelname aan deze pre-test. Mijn naam is Merel Kant, masterstudent Communication & Information Sciences aan Tilburg University. In het kader van mijn masterthesis doe ik onderzoek naar interacties op het online dating platform Tinder onder Nederlandse jongvolwassenen (18-34 jaar).

Tijdens dit onderzoek ga je kijken naar 18 verschillende openingszinnen. Alle antwoorden worden vertrouwelijk en anoniem verwerkt en je kunt op elk gewenst moment besluiten de enquête te beëindigen zonder nadelige gevolgen. Je gegevens zullen uitsluitend toegankelijk zijn voor personen die direct betrokken zijn bij dit onderzoek. Bij vragen kun je contact opnemen via:

m.kant@tilburguniversity.edu.

De survey duurt ongeveer 5 tot 10 minuten. Alvast bedankt voor je deelname!

- a. Ik ga akkoord en wil beginnen met de survey

Demographic Questions

1. Wat is je leeftijd?
(text entry response)
2. Wat is je genderidentiteit?
 - a. Man
 - b. Vrouw
 - c. Non-binair
 - d. Ik geef hier liever geen antwoord op

Evaluation Criteria

Geef op een schaal van 1 tot 7 aan wat je van deze openingszin vindt.

1. Niet humoristisch - Humoristisch
2. Niet flirterig - Flirterig
3. Indirect - Direct
4. Niet complimenteaus - Complimenteaus
5. Zonder bijbedoelingen – Met bijbedoelingen

Debriefing

Hartelijk dank voor je deelname! Jouw inbreng is van groot belang voor mijn onderzoek naar online dating. Mocht je vragen of opmerkingen hebben, dan kun je contact opnemen via:

m.kant@tilburguniversity.edu

Appendix B

Results Pre-test

Table 2 shows the results of the pre-test. Type refers to the type of pick-up line, with FL standing for Flippant, DC for Direct complimentary and IN for Innocuous. In addition, form refers to 'question or no question', with Q standing for Question and S for Statement. Participants rated the 18 pick-up lines on a 7-point Likert scale (1 = "*strongly disagree*," 7 = "*strongly agree*") regarding their perceived humor (HU), flirtatiousness (FL), directness (DI), complimentary nature (CN), and underlying intentions (UI). The relevant variables were added together per pick-up line and the average was calculated as the final score.

Table 2

Results of the pre-test.

Type	Form	Pick-up line	HU	FL	DI	CN	UI	Total
FL	S	Ik wist niet dat perfectie bestond totdat ik jou profiel zag.	3.73*	6.23*	5.82	6.18	5.64	4.98
FL	Q	Hoe voelt het om het beste profiel op deze app te hebben?	4.36*	5.73*	5.18	5.95	5.05	5.05
FL	S	Je zal wel moe zijn, want je loopt al de hele dag rondjes in mijn hoofd.	5.48*	5.52*	4.52	4.67	5.14	5.55
FL	Q	Ben je niet moe van al die rondjes die je in mijn hoofd loopt?	4.76*	5.38*	4.48	4.00	5.38	5.07
FL	S	Ik was niet van plan om hier verliefd te worden, maar jij verandert de zaak	4.76*	6.43*	6.19	6.05	5.33	5.60
FL	Q	Geloof je in liefde bij de eerste swipe, of moet ik mijn match ongedaan maken en het opnieuw proberen?	5.67*	5.76*	5.10	5.00	5.76	5.72
DC	Q	Wat een mooie glimlach, wat maakt jou blij?	2.65	4.17	4.78*	6.13*	3.22	5.46
DC	S	Je hebt een prachtige glimlach	1.95	5.27	5.41*	6.23*	3.91	5.82
DC	Q	Waar zijn die prachtige foto's in het buitenland genomen?	1.86	2.71	3.05*	4.38*	3.14	3.72
DC	S	Hey, wat een prachtige foto's heb je.	1.57	4.71	4.52*	6.10*	3.86	5.31
DC	Q	Wat is het beste compliment dat je ooit heb gekregen? Je krijgt ze vast vaak met zo'n leuke uitstraling.	3.19	5.71	4.57*	5.95*	5.14	5.26

DC	S	Hey, wat heb je een leuke uitstraling.	1.59	5.18	5.45*	6.23*	3.59	5.84
IN	Q	Hey, hou je ook van reizen?	1.64	1.82	3.59*	1.50	2.91*	3.21
IN	S	Hey, ik hou ook van reizen.	1.43	1.86	2.29*	1.81	2.95*	2.62
IN	Q	Hey, ken je nog een goede serie?	1.62	1.86	2.24*	1.48	2.81*	2.53
IN	S	Hey, ik kijk ook graag series.	1.64	1.68	2.50*	1.55	3.23*	2.87
IN	Q	Ben je een honden- of kattenmens?	2.32	1.73	2.64*	1.32	2.64*	2.64
IN	S	Ik ben een echte hondenliefhebber.	1.76	1.67	2.71*	1.38	2.62*	2.67

*Note: *Indicates the variables included in the total score. All measures range from 1 (strongly disagree) to 7 (strongly agree).*

Appendix C

Stimuli Materials

Figure 7

Screenshots of Tinder chats with direct complimentary questions.

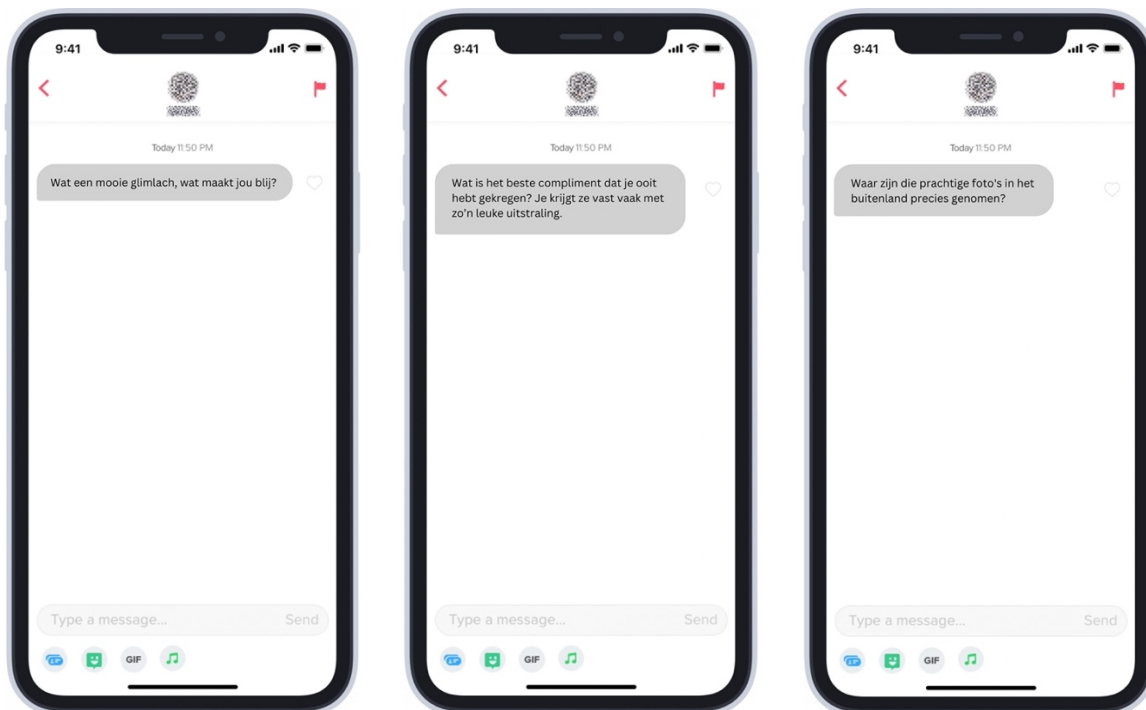


Figure 8

Screenshots of Tinder chats with direct complimentary statements.

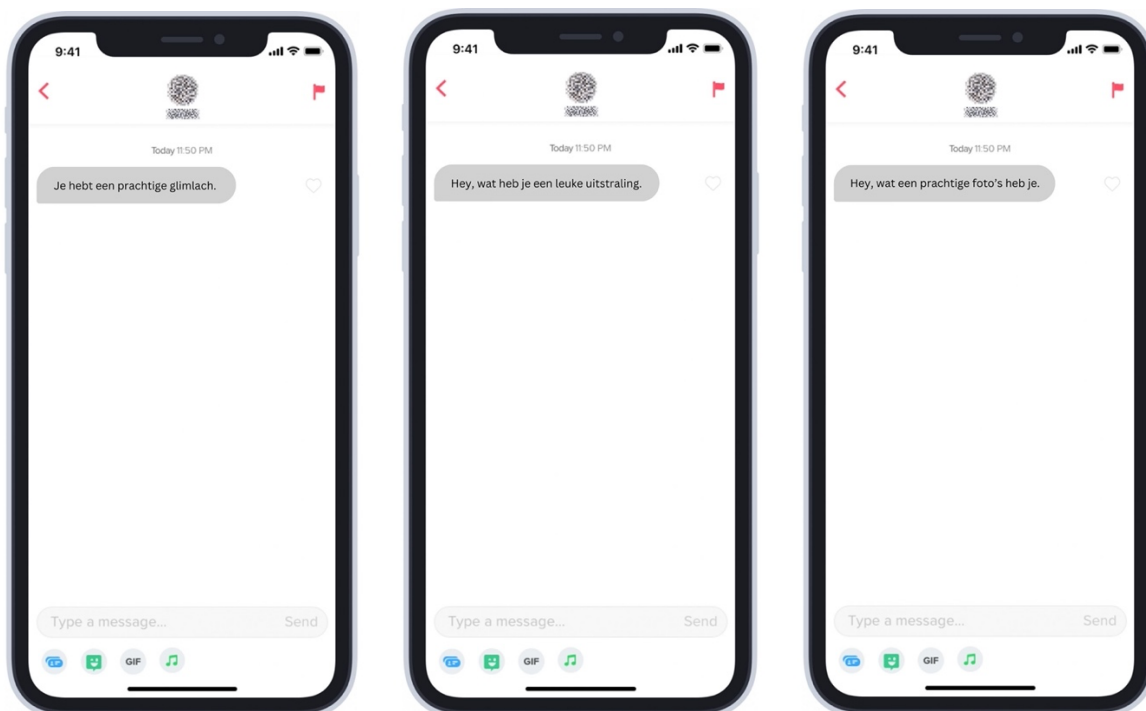
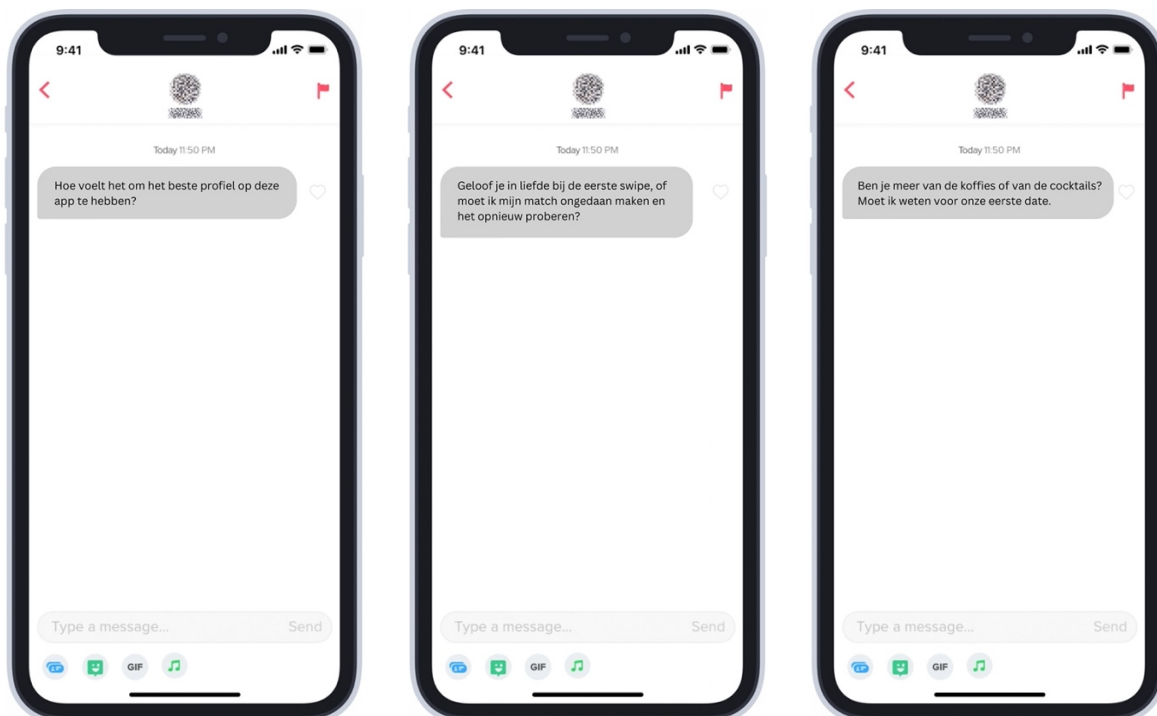


Figure 9

Screenshots of Tinder chats with flippant questions.

**Figure 10**

Screenshots of Tinder chats with flippant statements.

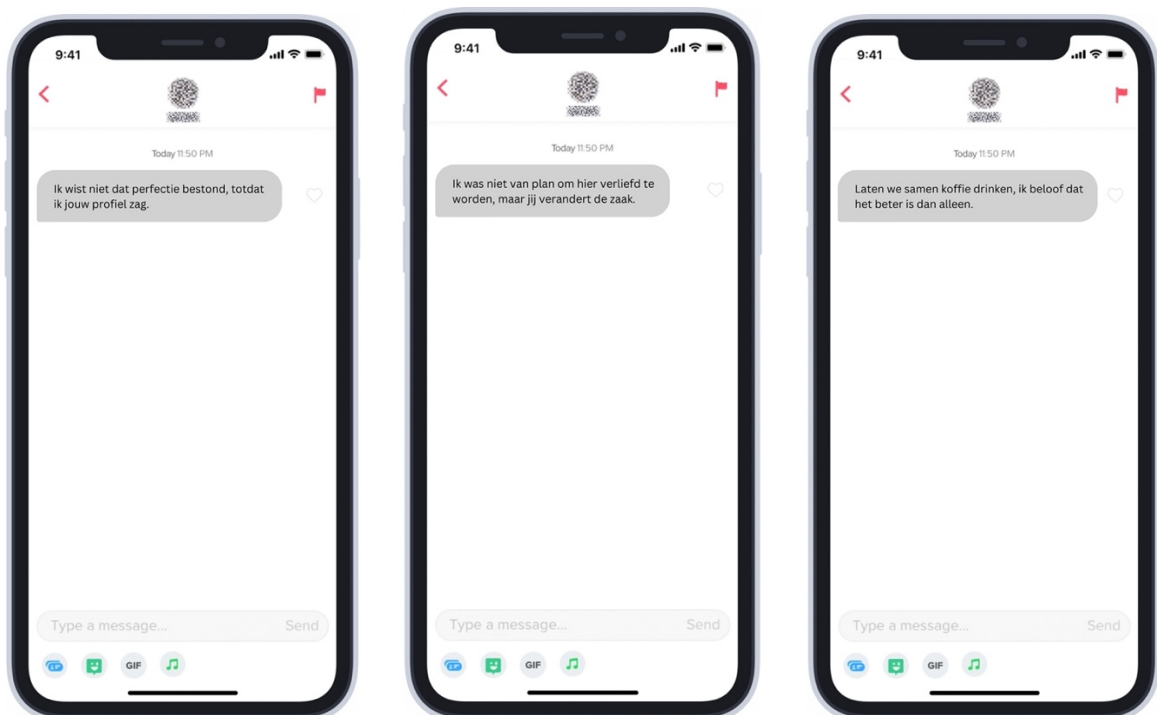
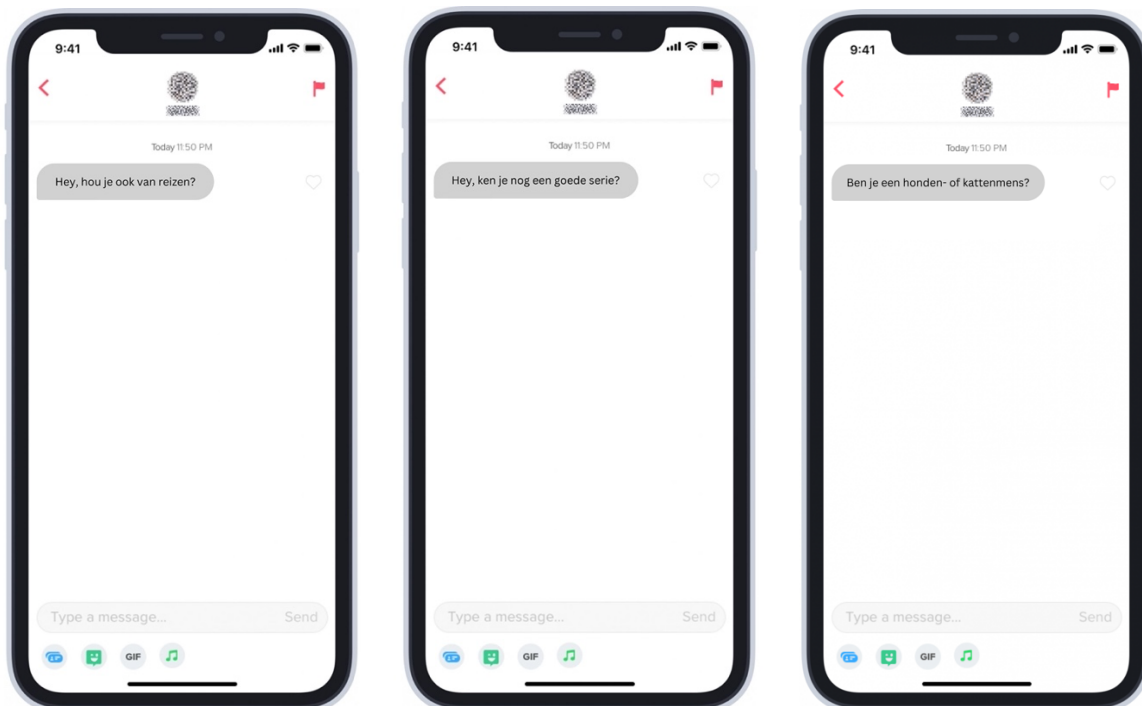
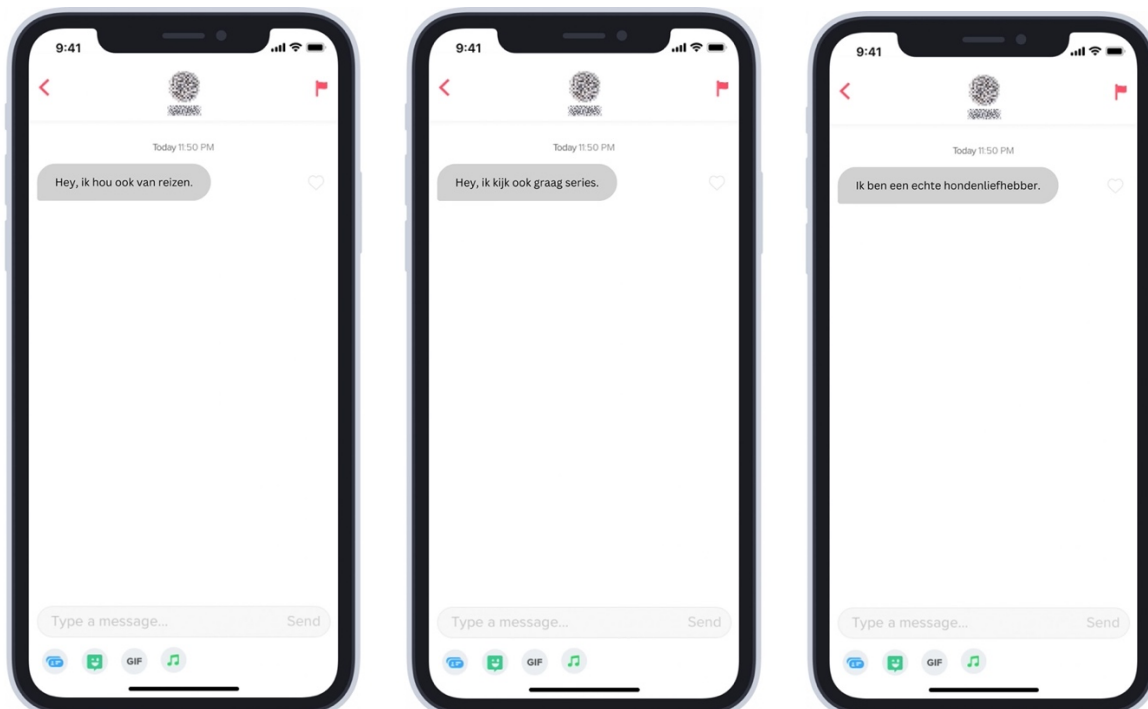


Figure 11

Screenshots of Tinder chats with innocuous questions.

**Figure 12**

Screenshots of Tinder chats with innocuous statements.



Appendix D

Survey

Instructions

Bedankt voor je deelname aan dit onderzoek. Mijn naam is Merel Kant, masterstudent Communication & Information Sciences aan Tilburg University. In het kader van mijn masterthesis doe ik onderzoek naar interacties op het online dating platform Tinder onder Nederlandse jongvolwassenen (18-34 jaar).

Tijdens dit onderzoek ga je kijken naar zes verschillende openingszinnen. Alle antwoorden worden vertrouwelijk en anoniem verwerkt en je kunt op elk gewenst moment besluiten de enquête te beëindigen zonder nadelige gevolgen. Je gegevens zullen uitsluitend toegankelijk zijn voor personen die direct betrokken zijn bij dit onderzoek. Bij vragen kun je contact opnemen via:

m.kant@tilburguniversity.edu.

De survey duurt ongeveer 5 tot 10 minuten.

Alvast bedankt voor je deelname!

- a. Ik ga akkoord en wil beginnen met de survey

Demographic Questions

1. Wanneer was je (ongeveer) voor het laatst actief op Tinder?
 - a. Minder dan een week geleden
 - b. Tussen een maand en zes maanden geleden
 - c. Meer dan zes maanden geleden
 - d. Ik heb nog nooit Tinder gebruikt
2. Wat is je leeftijd?
(text entry response)
3. Wat is je genderidentiteit?
 - a. Man
 - b. Vrouw
 - c. Non-binair
 - d. Ik geef hier liever geen antwoord op

Perception

Stel je voor dat je het volgende bericht ontvangt van een Tinder-match, die je afgelopen week voorbij zag komen en naar rechts hebt geswiped/geliked.

Geef aan in hoeverre je het eens bent met de volgende stellingen op een schaal van 1 tot 7, waarbij 1 = helemaal niet mee eens en 7 = helemaal mee eens:

Betrouwbaarheid

1. Ik denk dat deze persoon te vertrouwen is.
2. Ik denk dat deze persoon eerlijk is.
3. Ik denk dat deze persoon betrouwbaar is.

Intelligentie

4. Ik denk dat deze persoon slim is.
5. Ik denk dat deze persoon intelligent is.

Sociabiliteit

6. Ik denk dat deze persoon sociaal is.
7. Ik denk dat deze persoon vriendelijk is.
8. Ik denk dat deze persoon zelfverzekerd is.

Humor

9. Ik denk dat deze persoon humor heeft.
10. Ik denk dat deze persoon grappig is.

Creativiteit

11. Ik denk dat deze persoon creatief is.
12. Ik denk dat deze persoon origineel is.

Romantische aantrekkingskracht

13. Ik zie deze persoon als een geschikte datingpartner.
14. Ik zou positief over mezelf denken als ik met deze persoon zou daten.
15. Ik denk dat mijn vrienden positief over mij zouden denken als ik met deze persoon zou daten.

Intentie om te reageren

16. Ik zou antwoorden op het bericht van deze persoon.
17. Ik zou enthousiast zijn om op deze persoon te reageren.
18. Ik zou ernaar uitkijken om met deze persoon een online gesprek te voeren.

Debriefing

Hartelijk dank voor je deelname! Jouw inbreng is van groot belang voor mijn onderzoek naar online dating. Mocht je vragen of opmerkingen hebben, dan kun je contact opnemen via:

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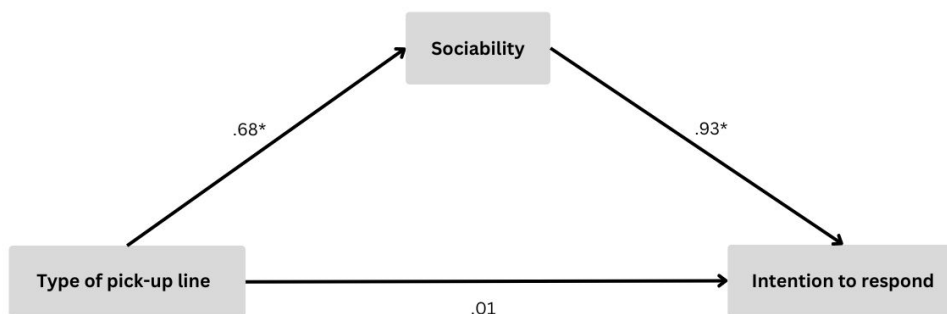
Appendix E

Full mediations visualized in graphs

Figures 13 and 14 illustrate the results of significant mediation analyses.

Figure 13

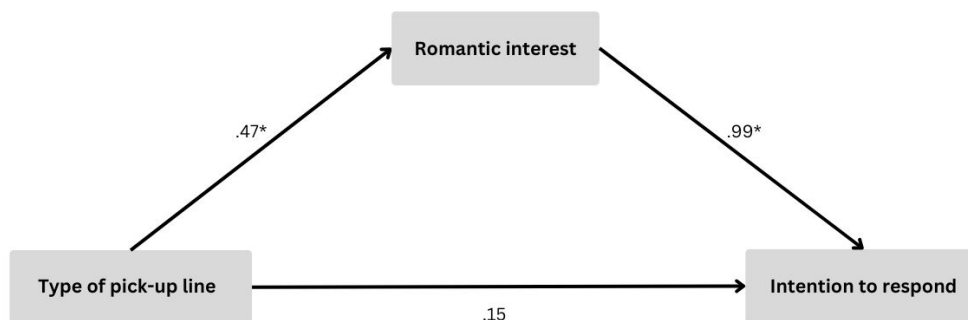
Results of the mediation analysis displaying perceived sociability significantly mediating the effect of type of pick-up line on intention to respond.



Note: * indicates $p < .05$

Figure 14

Results of the mediation analysis displaying perceived romantic interest significantly mediating the effect of type of pick-up line on intention to respond.



Note: * indicates $p < .05$